HUMANIZATION REVISION SUMMARY

BACSE202 – Database Systems Project Final Evaluation Report
Criminal Case Management & Crime Analytics System (CCMS)

Institution: VIT Chennai, School of Computer Science and Engineering (SCOPE)
Course: BACSE202 – DATABASE SYSTEMS | Semester: Fall / Winter 2026
Students: Ayush Sharma [25BAI1392] & Ojas Sugur [25BAI1222]
Supervisor: DR. HEMALATHA K (Assistant Professor Senior Grade 1)
Document Modified: Criminal_Case_Management_Final_Report_Humanized.docx
Date of Revision: 30/10/2026

1. Executive Overview & Humanization Methodology

This revision summary documents the thorough, in-place humanization of our academic project report. The objective was to transform formulaic, overly generic, or repetitive AI-like phrasing into clear, authentic academic English that reflects the genuine voice and hands-on understanding of B.Tech Artificial Intelligence and Machine Learning students who engineered this database system.

Crucially, the document editing was executed directly on a structured copy of the approved report (Criminal_Case_Management_Final_Report_Humanized.docx). All visual formatting, section breaks, Roman and decimal page numbering, table borders, embedded images (including the official VIT Chennai logo, EER diagram, and ERD diagram), code block tables, and references were preserved with zero structural disruption.

2. Chapter-by-Chapter Improvements & Concrete Implementation Linkages

Section / Chapter	Humanized Phrasing Improvements	DBMS & Real-World Concepts Integrated
Preliminary Pages (Abstract, Decl, Cert, Ack)	Rewrote Abstract to 315 natural words across 3 balanced paragraphs + 10 keywords. Polished Declaration, Certificate, and Acknowledgement with respectful, authentic student gratitude.	Explains project motivation, 14-table BCNF schema, Bell-LaPadula security, Citizen Portal, and 35 Pytest benchmarks without generic fluff.
Chapter 1: Problem Statement	Replaced generic buzzwords with operational realities of municipal police station desks (handwriting FIR dockets, lost papers, unnormalized spreadsheet columns).	Connects operational failures to relational anomalies (insertion, deletion, update) and tamperable custody logs lacking forensic audit trails.

Chapter 2: Project Objectives & Scope	Reframed objectives as student engineering goals rather than abstract corporate marketing claims.	Clarified why dual MySQL accounts were created, why BCNF was targeted, and how 50k Chicago records were reconciled via canonical IUCR lookups.
Chapter 3: Literature Survey & Foundations	Linked prior policing studies directly to our progression from DA1 (Ojas Sugur) to final implementation.	Grounds the system in classic database literature (Elmasri & Navathe for BCNF, Bell & LaPadula for MLS, Silberschatz for transaction controls).
Chapter 4: Requirements & Roles	Specified the 4 operational roles and 12 functional requirements using precise, unambiguous student wording.	Emphasizes password hashing with salted scrypt (32768:8:1) and anti-enumeration HTTP 404 security design for citizen reports.
Chapter 5: System Design & Architecture	Explained the three-tier design from the viewpoint of student developers implementing Flask and MySQL.	Articulates why InnoDB was chosen (ACID transactions, foreign key cascades, virtual columns) and how Flask injects session variables (@app_user_id) into MySQL.
Chapter 6: Database Design, ER & EER	Narrated the authentic evolution from DA1's initial 6-table concept to the 14-table BCNF enterprise architecture.	Explains the Enhanced ER Disjoint Total Specialization (d, t) using a single table with ENUM discriminator to avoid expensive outer joins.
Chapter 7: Schema & Data Dictionary	Preserved formal relational notation and comprehensive 14-table Data Dictionary exactly as defined.	Maintains all primary keys [PK], candidate keys [CK], foreign keys, and column types without altering technical names.
Chapter 8: Normalization Proofs	Clarified the mathematical proofs for USERS, CASES, and CASE_ASSIGNMENTS in accessible academic	Explains the virtual generated column active_assignment_key and how its UNIQUE index

	language.	mathematically enforces BCNF without concurrency races.
Chapter 9: SQL Implementation Catalog	Connected DA2 foundational SQL queries (Ayush Sharma) to the expanded production queries for the web platform.	Explains multi-table joins, subqueries, grouping aggregates, and analytical views (v_case_summary).
Chapter 10: Advanced Database Features	Detailed server-side database routines (sp_assign_case, fn_case_duration, triggers, cursors) from a practical implementation standpoint.	Explains atomic investigator reassignment inside an ACID transaction and tamper-evident custody logging using session variables.
Chapter 11: Cloud Deployment Architecture	Distinguished between verified local development (Python 3.14 + MySQL 9.7) and planned production cloud topology.	Describes multi-stage Docker containerization, Application Load Balancers, private subnets, and secrets management.
Chapter 12: Testing & Quality Assurance	Documented test methodology with authentic student rigor, citing actual test scripts and outcomes.	Reports the 35 automated Pytest integration tests and confirms MySQL Error 1142 ('command denied') on least-privilege violation attempts.
Chapter 13: Performance Analysis & Indexing	Explained how importing 50,000 Chicago incidents necessitated secondary B-tree composite indexing.	Analyzes MySQL EXPLAIN plans, demonstrating how idx_pci_district shifted access from ALL to range/ref, cutting scanned rows by >95% and latency from 160ms to <5ms.
Chapter 14: Results & Discussion	Documented genuine development hurdles, design tradeoffs, and technical lessons learned.	Discusses least-privilege unit test errors resolved with UUID emails, anti-enumeration 403 vs 404 security decisions, and null-

		safe template guards.
Chapter 15: Conclusion	Summarized the project trajectory from DA1 conceptual sketches to a fully verified 14-table BCNF database system.	Emphasizes the value of enforcing business constraints and security policies directly inside the database engine.
Chapter 16: Future Scope	Articulated practical future extensions in clear, realistic terms.	Covers MySQL spatial GIS extensions (POINT, R-tree indexes), biometric vector similarity search, and blockchain custody anchoring.

3. Verification of Preserved Formatting & Visual Elements

Visual / Structural Element	Status	Verification Note
Official VIT Chennai Logo	PRESERVED	Present on Cover Page (width=3.8 in) and Submission Page (width=3.3 in) from docs/vit_logo.png.
Cover Page Design	PRESERVED	BACSE202, title, video link placeholder, VIT logo, institution, and date October 2026 intact.
Submission & Preliminary Pages	PRESERVED	Title page, Declaration, Certificate, Abstract, Acknowledgement, TOC, LOT, LOF, and Abbreviations present.
Team Details & Reg Numbers	PRESERVED	Ayush Sharma [25BAI1392] and Ojas Sugur [25BAI1222] verified across all pages.
Tables & Data Dictionaries	PRESERVED	All 26 tables (including 14-table Data Dictionary and code block tables) remain intact.

Diagrams & Illustrations	PRESERVED	EER Specialization diagram and Complete 14-Table Relational ERD diagram embedded and captioned.
Section Breaks & Numbering	PRESERVED	Section 0 unnumbered, Section 1 lowerRoman (i-x), Section 2 decimal Arabic (1-37) starting at page 1.
PDF Export & Page Inspection	VERIFIED	Compiled via native Microsoft Word COM automation to 47 clean, unbroken pages (586 KB).

4. Factual Details Flagged for Final Student / Faculty Confirmation

The following factual placeholders and operational parameters have been maintained transparently in the document and can be customized before final evaluation submission:

1. Project Explanation Video Link: Located on the cover page as 'Project Explanation Video: [INSERT VIDEO LINK]'. Please insert your recorded YouTube/Google Drive demonstration link before final printing.

2. Faculty Supervisor Designation: Listed as 'DR. HEMALATHA K (Assistant Professor Senior Grade 1)'. If an alternate project guide or co-guide is assigned, update the placeholder on the Declaration and Certificate pages.

3. Date and Submission Place: Recorded as 'Place: VIT CHENNAI' and 'Date: 30/10/2026'. Adjust the submission date if your evaluation or viva occurs on a subsequent schedule.

4. Live Test Suite Execution: Confirmed as 35 passing Pytest integration tests and a 24-point smoke test suite executed directly against MySQL Community Server 9.7.1 on Windows.